

"Modules manufactured today are more horizontal-specific rather than vertical-specific. Of course, there are some modules manufactured specifically for particular verticals but these are more expensive."

Sogani adds, "There are very high-output power modules that can go up to 25dBm or modules catering to vehicle-to-vehicle communication; such modules are meant for niche applications." Manufacturers are focusing on creating solutions with a high level of technology and software integration that can be used across multiple verticals.

New features for IoT

Redpine Signals recently announced a Wi-Fi, Bluetooth and ZigBee combination module, which is very unique in the market. Bluetooth includes Bluetooth Classic and BLE. Sogani says, "There is no other vendor who has this combination. With such a module, it is possible to create a system that is a perfect gateway for different wireless standards."

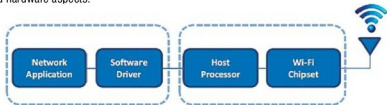
He adds, "Data can be collected on Bluetooth, ZigBee and Wi-Fi, and transferred to the cloud using Wi-Fi. We are seeing that Wi-Fi is becoming the hub for other wireless technologies to get onto the Internet." With this unique combination of multiple wireless standards—Wi-Fi, Bluetooth and ZigBee—in one module, area, power and cost get reduced significantly, making it ideal for wearable devices.

With the advent of the IoT, everyone is jumping onto the IoT bandwagon with all they have, feels T. Anand, managing director, Knewron. Atmel launched WILC3000/WINC1500 series of Wi-Fi solutions in module format for developers. He says, "These modules are highly powerful and can be controlled via multiple types of serial interfaces."

"Additionally, another new silicon vendor, Espressif, has also jumped in with their newly-launched ESP8266 systems on chip (SoC) as well as Wi-Fi modules around it. These modules are extremely cheap as compared to

How to select the right Wi-Fi module for your application

When choosing the Wi-Fi module for your applications, you need to consider both software and hardware aspects.



A simplified block diagram of a Wi-Fi module

Hardware of the Wi-Fi module contains two main parts: a Wi-Fi chip and an application host processor. For more standalone Wi-Fi applications, engineers can select from Wi-Fi modules from makers like Laird Technologies, Quatech and Roving Networks that combine the signal chain with a complete embedded processor core. The application host processor has internal or external flash, read only memory (ROM) and random access memory (RAM). A number of I/Os are available to allow a wide range of applications. These include timers, serial communication interfaces, analogue comparators, analogue-to-digital converters ADCs, digital-to-analogue converters (DACs), crystal oscillators and a debug interface.

The specialised software is required by some applications to deliver secure, trouble-free operations and manageability that customers require. The Wi-Fi software usually includes a device driver, an integrated 802.11 security supplicant and a full-featured management and monitoring utility.

—Abhishek Rao, technical marketing manager, element14

all other Wi-Fi modules in the global market," Anand adds.

Shawn Hymel, creative engineer, SparkFun Electronics, informs that companies such as TI and Atmel are creating smaller, more integrated and lower-power Wi-Fi solutions. He says, "Atmel ATSAMW25, for example, contains an integrated ARM Cortex-M0+ processor along with a full 802.11 b/g/n stack. Other chips offer both Bluetooth and Wi-Fi on a single chip."

He adds, "These modules are being used to create remote sensors the size of a fingernail that can post data to Web services like Xively, AT&T's M2X, IBM's Bluemix, data.sparkfun and so on."

ESP8266 based modules are mainly universal asynchronous receiver/transmitter (UART) based modules and can act as access points by themselves, along with a standard Wi-Fi module, informs Anand. He says, "Espressif has also released a software development kit (SDK) for the chip, so that one can actually fully unleash the potential of the chip and use all other general-purpose inputs/outputs (GPIOs) of the chip/module."

He adds, "You probably would

not need another microcontroller for your small IoT applications as the SoC would give you one analogue-to-digital converter (ADC) and eight to ten GPIOs—all being pulse-width modulation (PWM) and interrupt capable."

Being in the market for over seven years, there have been significant advancements in Microchip's module offerings too. Abhinav Venuturumilli, Wi-Fi product line manager, WPD Marketing, Microchip Technology Inc., says, "The new additions to our portfolio in the past year include BLE module RN4020 and Bluetooth dual-mode module BM77 that has very small footprint and supports Bluetooth 4.0."

He adds, "Microchip has also released a sub-GHz module based on the new LoRaTM technology, enabling long-range coverage for low-power embedded applications." This RN2483 433MHz/868MHz module is ideal for battery-operated sensors and low-power applications such as the IoT, machine-to-machine (M2M), smart-city, sensor networks and industrial automation.

CC3200, a Wi-Fi SoC with an integrated controller with video and audio